

Thermal Radiation

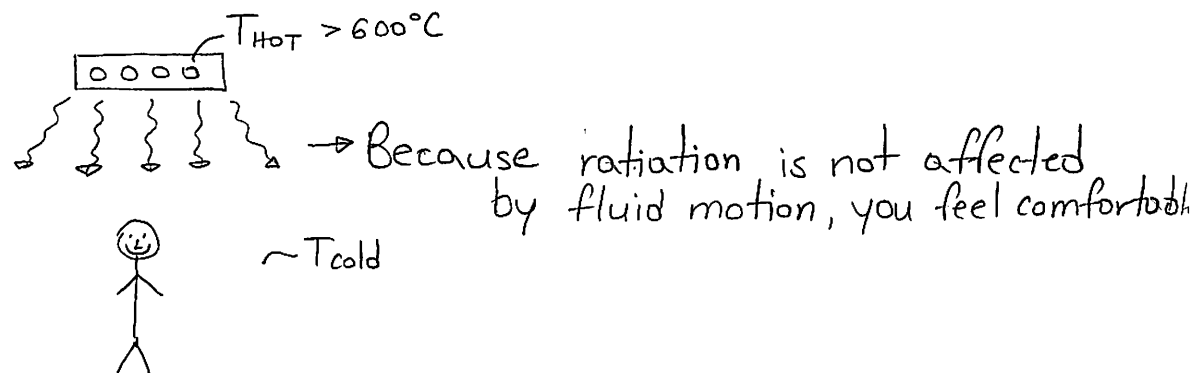
So far in ME 320, we've dealt with the following:

Conduction } Heat transfer by a temperature gradient in
Convection } a medium.

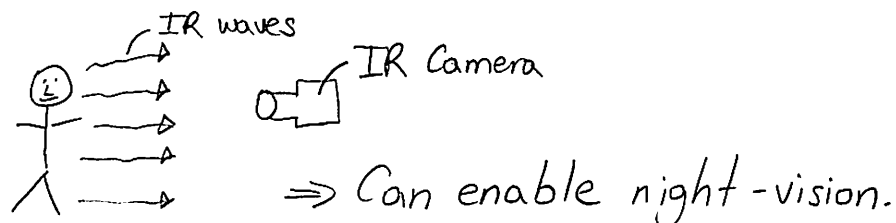
Radiation } - no medium required (can propagate in vacuum)
 } - a form of energy emitted by all matter
 } at a finite temperature.

Applications & Examples:

- 1) Global Energy Balance $\rightarrow$ Earth receives its energy from the sun.
- 2) HVAC $\rightarrow$ Heating, Ventilation & Air Conditioning
 $\hookrightarrow$ Radiative heaters (bar or patio heaters)



3) Infrared Detection



Theory:

① Electromagnetic Theory (Maxwell) $\Rightarrow$ Light is a wave

$$c = \lambda \nu = \frac{3.0 \times 10^8 \text{ m/s}}{n}$$

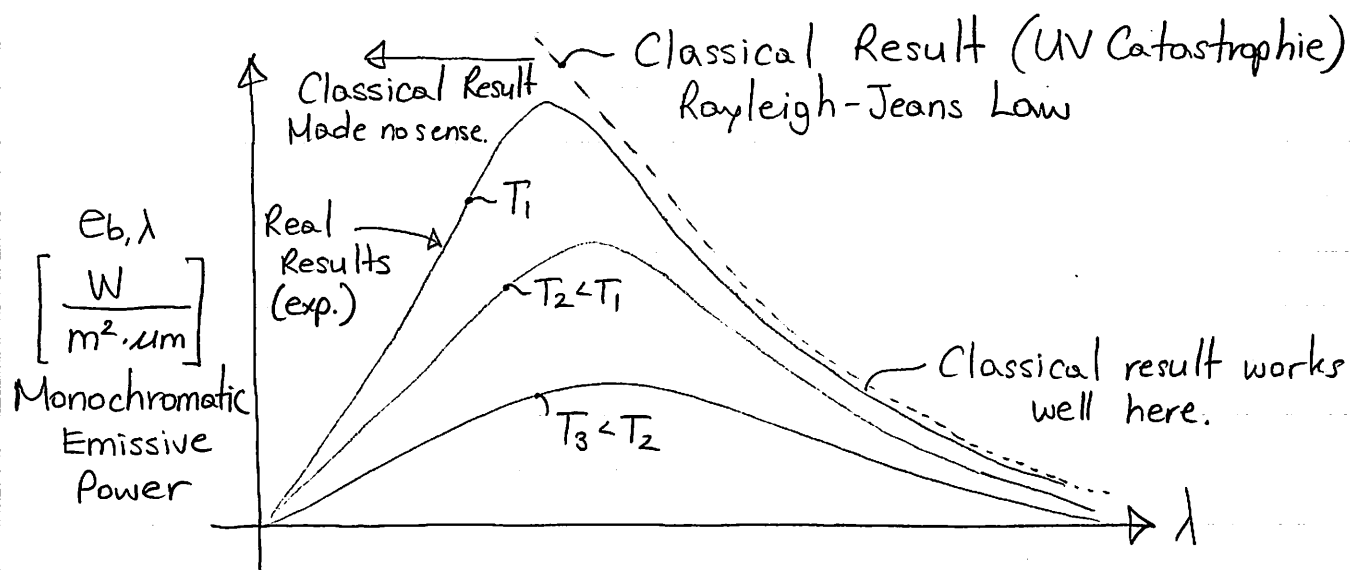
$c \equiv$ speed of light
 $n \equiv$ medium index of refraction
 $\nu = \frac{c}{\lambda} \equiv$ frequency of light
 $\lambda \equiv$ wavelength of light

② Quantum Theory (Planck) $\Rightarrow$ Light is a particle (photon)

$$e = \frac{hc}{\lambda} = h\nu$$

$e \equiv$ energy of a photon [J]
 $c \equiv$ speed of light in a medium
 $\lambda \equiv$ wavelength of light
 $h \equiv 6.6256 \times 10^{-34} \text{ J}\cdot\text{s}$ (Planck's const.)

These two theories came to a head in early 1900's.
Classical theory didn't make sense:



To solve this problem, Planck assumed light was made of particles, and applied kinetic theory. This was the basis for quantum theory.

By treating light as particles & using statistical thermody.
he derived:

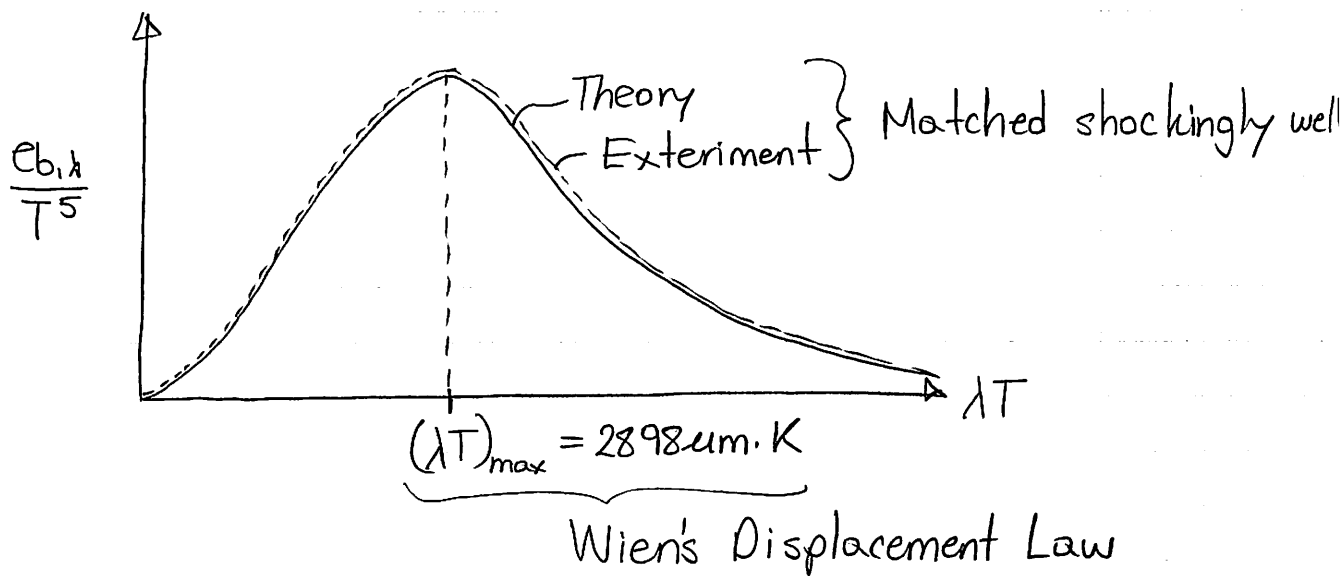
$$e_{b,\lambda} = \frac{2\pi h c^2 \lambda^{-5}}{e^{\frac{hc}{k_B \lambda T}} - 1} = \frac{C_1 \lambda^{-5}}{e^{\frac{C_2}{\lambda T}} - 1}$$

Dividing both sides by T^5 :

$$\boxed{\frac{e_{b,\lambda}}{T^5} = \frac{C_1 (\lambda T)^{-5}}{e^{\frac{C_2}{\lambda T}} - 1}}$$

$k_B \equiv$ Boltzman constant
 $C_1 = 2\pi h c^2 = 3.742 \times 10^8 \frac{W \mu m^4}{m^2}$

$C_2 = \frac{hc}{k_B} = 1.4389 \times 10^4 \mu m \cdot K$



Visible wavelength range: $0.4 < \lambda < 0.7 \mu m$ (blue to red)
 Thermal radiation: $0.1 - 100 \mu m$

So as energy increases, $(T \uparrow)$, the peak emissive wavelength is shifted to lower wavelengths (visible)

Example: Hot IRON Bar $\Rightarrow$ goes from no color $\rightarrow$ red $\rightarrow$ orange $\rightarrow$ yellow $\rightarrow$ white (158)